Republic of the Philippines
University of Cabuyao
(Pamantasan ng cabuyao)
COLLEGE OF COMPUTING STUDIES
Katapatan Mutual Homes, Brgy. Banay-banay, City of Cabuyao, Laguna 4025

Student Name (LN, FN MI.):				**Date:**	
Section:		**Instructor:**	Ms. Sairine C. Pregonero	**Sem./ A.Y.**	2nd/25-26

I. Correlation Analysis

A teacher wants to determine if there is a relationship between study hours and exam scores of students.

Data

Student	Study Hours (x)	Exam Score (y)
1	2	65
2	4	70
3	6	75
4	8	85
5	10	90

Question 1: Compute the Pearson Correlation Coefficient (r)

Question 2: Interpret the Relationship

Question 3: Test of Significance (a = 0.05)

II. Regression Analysis

A company recorded advertising expenses and sales revenue.

Data

Month	Advertising Expense (x)	Sales Revenue (y)
1	5	50
2	6	55
3	7	60

4	8	65
5	9	70

Question 1: Find the Regression Equation $y = a + bx$

Question 2: Predict Sales Revenue when x = 12

Question 3: Interpret the Slope

III. One-Way ANOVA

A professor wants to know if there is a significant difference in test scores among three teaching methods.

Data

Method A	Method B	Method C
78	85	90
80	88	92
82	86	91

Question 1: State the Hypotheses

- H_0:
- H_1:

Question 2: Complete the ANOVA Table

Source of Variation	SS	df	MS	F
Between (Methods)				

Within (Error)				
Total				

Question 3: Test at a = 0.05

F-computed: _____ F-critical: _____

_____

Question 4: State the Conclusion

IV. Two-Way ANOVA

A researcher studies the effect of teaching method and subject area on student performance.

Data

Subject	Method A	Method B	Method C
Math	85	88	90
English	80	84	86
Science	82	85	89

Question 1: Test - Significant Difference Among Teaching Methods

- H_0:
- H_1:

Question 2: Test - Significant Difference Among Subjects

- H_0:
- H_1:

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Question 3: Complete the ANOVA Table

Source of Variation	SS	df	MS	F
Methods (Columns)				
Subjects (Rows)				
Error (Interaction)				
Total				

Question 4: Interpret the Result